WEEK – 10

1# A Mobile Phone Service Center wants to maintain a database to manage information about customers, mobile phones, and service requests. Currently, records are maintained manually, which causes data inconsistency and difficulty in retrieval. You are required to design and implement a database system for the service center.

a) Create the following tables based on the given scenario:

1. Customer

- Customer_ID
- Customer_Name
- Mobile_No
- Address

```sql
CREATE TABLE Customer1 (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(50),
    Mobile_No VARCHAR(15),
    Address VARCHAR(100)
);
```

2. Mobile

- Mobile_ID
- Brand
- Model
- Customer_ID

```sql
CREATE TABLE Mobile (
    Mobile_ID INT PRIMARY KEY,
    Brand VARCHAR(50),
    Model VARCHAR(50),
    Customer_ID INT,
    FOREIGN KEY (Customer_ID) REFERENCES Customer1(Customer_ID)
);
```

3. Service_Request

- Service_ID
- Mobile_ID
- Problem_Description
- Service_Date
- Service_Cost

```sql
CREATE TABLE Service_Request (
    Service_ID INT PRIMARY KEY,
    Mobile_ID INT,
    Problem_Description VARCHAR(100),
    Service_Date DATE,
    Service_Cost DECIMAL(10,2),
    FOREIGN KEY (Mobile_ID) REFERENCES Mobile(Mobile_ID)
);
```

(Choose appropriate data types and define primary keys and foreign keys where required.)

b) Insert at least five records into each table using SQL INSERT statements.

```sql
INSERT INTO Customer1 VALUES
(1, 'Amit Sharma', '9876543210', 'Aligarh'),
(2, 'Ravi Kumar', '9123456780', 'Delhi'),
(3, 'Sita Verma', '9988776655', 'Agra'),
(4, 'Anil Gupta', '9090909090', 'Kanpur'),
(5, 'Neha Singh', '8888888888', 'Lucknow');

INSERT INTO Mobile VALUES
(101, 'Samsung', 'Galaxy S21', 1),
(102, 'Apple', 'iPhone 12', 2),
(103, 'Xiaomi', 'Redmi Note 10', 3),
(104, 'OnePlus', 'Nord CE', 4),
(105, 'Vivo', 'V20', 5);

INSERT INTO Service_Request VALUES
(201, 101, 'Screen Damage', TO_DATE('2025-01-10', 'YYYY-MM-DD'), 3000),
(202, 102, 'Battery Issue', TO_DATE('2025-01-12', 'YYYY-MM-DD'), 2000),
(203, 103, 'Software Problem', TO_DATE('2025-01-15', 'YYYY-MM-DD'), 1500),
(204, 104, 'Charging Issue', TO_DATE('2025-01-18', 'YYYY-MM-DD'), 1800),
(205, 105, 'Camera Problem', TO_DATE('2025-01-20', 'YYYY-MM-DD'), 2200);
```

Results | Explain | Describe | Saved SQL | History

```
5 row(s) inserted.
```

c) Display all customer details.

```
SELECT * FROM Customer;
```

Results | Explain | Describe | Saved SQL | History

CUSTOMER_NO	CUSTOMER_NAME	CUSTOMER_STREET	CUSTOMER_CITY
C1	Amit	Street1	Aligarh
C2	Rahul	Street2	Delhi
C3	Sneha	Street3	Mumbai
C4	Priya	Street4	Chennai
C5	Karan	Street5	Aligarh

5 rows returned in 0.03 seconds Download

d) Display all mobile phones serviced in the service center.

```
SELECT m.Mobile_ID, m.Brand, m.Model
FROM Mobile m
JOIN Service_Request s
ON m.Mobile_ID = s.Mobile_ID;
```

Results | Explain | Describe | Saved SQL | History

MOBILE_ID	BRAND	MODEL
101	Samsung	Galaxy S21
102	Apple	iPhone 12
103	Xiaomi	Redmi Note 10
104	OnePlus	Nord CE
105	Vivo	V20

5 rows returned in 0.04 seconds Download

e) Display customer names along with their mobile phone details.

```
SELECT c.Customer_Name, m.Brand, m.Model
FROM Customer1 c
JOIN Mobile m
ON c.Customer_ID = m.Customer_ID;
```

Results | Explain | Describe | Saved SQL | History

CUSTOMER_NAME	BRAND	MODEL
Amit Sharma	Samsung	Galaxy S21
Ravi Kumar	Apple	iPhone 12
Sita Verma	Xiaomi	Redmi Note 10
Anil Gupta	OnePlus	Nord CE
Neha Singh	Vivo	V20

5 rows returned in 0.03 seconds Download

f) Display service details for a particular mobile phone.

```sql
SELECT m.Brand, m.Model, s.Problem_Description, s.Service_Date, s.Service_Cost
FROM Service_Request s
JOIN Mobile m
ON s.Mobile_ID = m.Mobile_ID
WHERE m.Mobile_ID = 101;
```

Results | Explain | Describe | Saved SQL | History

BRAND	MODEL	PROBLEM_DESCRIPTION	SERVICE_DATE	SERVICE_COST
Samsung	Galaxy S21	Screen Damage	1/10/2025	3000

1 rows returned in 0.03 seconds Download